

Assignment: Express.js REST API with MySQL

Assignment Title

Express.js Users API using MySQL with Modular Folder Structure

Objective

The objective of this assignment is to assess the learner's understanding of:

- Express.js application structure
 - REST API design using GET and POST methods
 - MySQL database connectivity
 - Middleware usage in Express
 - Separation of concerns using routes, database, and middleware folders
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Problem Statement

You are required to develop a **Users Management REST API** using **Express.js** and **MySQL**.

The application must follow a **proper Express folder structure** with:

- Separate routes folder
- Separate database configuration folder
- Use of middleware

The API should allow:

1. Fetching all users from the database
 2. Adding a new user into the database
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Functional Requirements

1. Database Requirements

- Create a MySQL database named:

express_db

- Create a table named:

users

- The **users** table must contain the following fields:

Column Name	Data Type	Description
id	INT (Primary Key, Auto Increment)	Unique user ID
first_name	VARCHAR(50)	User first name
last_name	VARCHAR(50)	User last name
email	VARCHAR(100)	User email
phone	VARCHAR(15)	User phone number

2. API Requirements

You must design **two REST APIs only**:

API 1: Get All Users

- HTTP Method: **GET**
- Endpoint: **/users**
- Description: Fetch and return all user records from the database

API 2: Add New User

- HTTP Method: **POST**
- Endpoint: **/users**
- Description: Insert a new user record into the database

3. Middleware Requirements

- Use built-in Express middleware to:
 - Parse JSON request bodies
- Create at least **one custom middleware**, such as:
 - Request logger (method and URL)
 - Validation middleware